Special Notice (SN) DARPA-SN-26-58: Organoid Cytomorphic Intelligence Resulting from Convergent Understanding and Information Transfer (O-Circuit) Proposers Workshop

DATE: April 10, 2026
LOCATION: 4075 Wilson Blvd, Suite 300, Arlington, VA 22203
REGISTRATION WEBSITE: https://events.sa-meetings.com/O-CircuitProposersWorkshop/
REGISTRATION DEADLINE: **Friday, April 3, 2026, at 4:00 PM EST** or when capacity is reached, whichever comes first.
TECHNICAL POC: Dr. Jeffrey Zaleski, DARPA/BTO
Email: O-Circuit@darpa.mil

The Biological Technologies Office (BTO) of the Defense Advanced Research Projects Agency (DARPA) is hosting a Proposers Workshop in support of an anticipated Program Solicitation (PS) for the Organoid Cytomorphic Intelligence Resulting from Convergent Understanding and Information Transfer (O-Circuit) program. The Proposers Workshop will be held on **Friday, April 10, 2026**, at the Executive Conference Center in Arlington, VA (4075 Wilson Blvd, Suite 300). A virtual attendance option will also be available. **Advance registration is required.** Registrants will have the opportunity to provide teaming profiles and give a lightning talk. There are a limited number of timeslots for lightning talks, so they must be scheduled upon registration on a first-come, first-served basis. Note, all times listed in this announcement and on the registration website are Eastern Time.

The goals of the Proposers Workshop are to:

- Support ideation of approaches to address the goals of the O-Circuit program
- Increase the efficiency of proposal refinement and preparation
- Encourage and promote teaming arrangements among organizations with the expertise, research facilities, and capabilities needed to execute research and development responsive to program goals.

DARPA anticipates releasing the O-Circuit Program Solicitation (PS) in early April 2026. When released, the PS will be available on https://sam.gov/. To maximize the pool of innovative proposal concepts, DARPA strongly encourages participation in the Proposers Workshop and related solicitations by non-traditional performers, including small and medium-sized businesses, as well as academic and research institutions, including first-time government contractors. While attendance at the Workshop is not mandatory for submitting a proposal or being selected for funding, it provides valuable insights and guidance. Information shared during the Proposers Workshop will be accessible on the O-Circuit Program page:

https://www.darpa.mil/research/programs/O-Circuit

The Proposers Workshop will include brief introductions by government personnel to the program concept and to working with DARPA. These will be followed by opportunities for potential proposers to highlight their technical capabilities and interests in teaming through lightening talks and a networking session (see below). DARPA will also collect participants' questions throughout the event and publish answers to a publicly available Frequently Asked Questions (FAQ) document.

Proposers Workshop Participation:

This Special Notice for Proposers Workshop includes the following opportunities for prospective proposers to showcase their capabilities and identify prospective team members:

1. *Lightning Talks.* Interested attendees are invited to submit one (1) Microsoft PowerPoint or Adobe PDF slide that summarizes their interests and capabilities using the template provided with this notice (**Attachment 1**) and on the registration website. The slides will be presented during 3-minute "lightning talks" at the O-Circuit Proposers Workshop. Submitted presentations consisting of multiple slides or a single slide with multiple layers will not be granted a time slot. Lightning talk submissions (no more than one per organization) will be accepted on a first-come, first-served basis, until the maximum possible number of submissions (given time constraints) is reached. Interested attendees must identify their intention to submit the lightning talk during registration. Due to limited availability, DARPA does not guarantee that these requests will be fulfilled. Following the lightning talks there will be an opportunity for in-person attendees to engage with other attendees and ideate on possible teaming strategies.

2. *Networking Session.* To help facilitate teaming interactions, during the registration process attendees will have the opportunity to select areas of expertise in which they see themselves as well-positioned to contribute and select areas in which they want to learn more (e.g., olfactory receptors/biosensing; input/output interfacing; cell culture health/composition). Dedicated time during the workshop will be provided for attendees to organize into groups where experts in one area interact with experts in the area(s) in which they have an interest in teaming with.

3. *Teaming Profiles.* Interested parties are also invited to submit a one- to two-page 'teaming profile' that includes contact information, program-relevant technical competencies, unique facilities, and/or desired competencies sought from other potential team partners. The profile should conform to the template provided with this notice (**Attachment 2**) and on the registration website.

   Profiles that do not use the provided Microsoft Word template (**Attachment 2**) and/or exceed the two-page limit will not be accepted. Information contained in teaming profiles shall be publicly releasable – profile submitters consent to distribution amongst event registrants. After Proposers Workshop, the teaming profiles will be sent via e-mail to all registrants. Specific content, communications, networking, and team formation are the sole responsibility of participants. Neither DARPA nor the Department of War (DoW) endorses any participating organization, nor does DARPA or DoW exercise any responsibility for improper dissemination of the teaming profiles.

All conforming lightning talks and teaming profiles must be emailed to O-Circuit@darpa.mil no later than **Monday, April 6, 2026, at 4:00 PM EST**. Specific content, communications, networking, and team formation are the sole responsibility of the participants.

Registration Information:

PLEASE NOTE: Registration closes Friday, April 3, 2026, at 4:00 PM EST

Participants must register in advance through the website. Registration is free but in-person attendance is limited by the venue capacity, so early registration is recommended. Interested parties are encouraged to coordinate attendance within their organizations prior to registration. In-person attendance is limited to no more than three representatives per division/department.

Interested parties are highly encouraged to have their technical PIs as well as a Contract representative participate. DARPA will have a dedicated 'ask me anything' panel related to the associated contractual and Other Transaction (OT)-related requirements and intricacies of working with DARPA on the O-Circuit program. Individuals who are unable to register because the deadline has occurred or capacity has been reached will be added to a waitlist. Slots that remain open after registration closes or that become available due to cancellations will be filled on a first-come, first-served basis from the waitlist. An online registration form and various other meeting details can be found at the registration website:

https://events.sa-meetings.com/O-CircuitProposersWorkshop/

All U.S. Permanent Residents and Foreign Nationals must submit a DARPA Form 60, except foreign government personnel who must submit only an Official Visit Request completed by their respective Embassy based in Washington, DC. All forms must be submitted no later than **<u>4:00 PM EST on April 3, 2025</u>**. Form 60 submission instructions are provided on the registration website and in the registration confirmation email. Contact your Embassy staff for assistance in submitting the Official Visit Request.

Further administrative questions should be addressed to O-Circuit@darpa.mil. Please refer to the O-Circuit Proposers Workshop (DARPA-SN-26-58) in all correspondence. This announcement is not a request for abstracts or proposals; any sent will be disregarded.

DARPA hosts Proposers Workshops to provide potential performers with a venue for interacting to facilitate teaming and to increase efficiency in proposal preparation and evaluation. Therefore, Proposers Workshops are open only to potential proposers who have registered.

<u>This notice is issued solely for information and potential new program planning purposes; the SN does not constitute a formal solicitation for proposals or proposal abstracts. In accordance with FAR 15.201(e), responses to this notice are not offers and cannot be accepted by the Government to form a binding contract. Submission is voluntary and is not required to propose to subsequent Broad Agency Announcements (if any) or research solicitations (if any) on this topic. DARPA will not provide reimbursement for costs incurred in responding to this SN. Respondents are advised that DARPA is under no obligation to acknowledge receipt of the information received or provide feedback to respondents with respect to any information submitted under this SN.</u>

<u>NO CLASSIFIED INFORMATION SHOULD BE INCLUDED IN THE SN RESPONSE.</u>

TENTATIVE PROGRAM INFORMATION

The program information provided below is from a **DRAFT** excerpt of the O-Circuit program solicitation. The information listed below (and throughout) is tentative **and subject to change** when the final solicitation is posted. **The details outlined in the official program solicitation will take precedence over any information contained in this special notice.**

Background and Objective

While semiconductor-based computation is readily deployed in well-resourced environments, it is challenged by sustained applications 'at the edge' due to inherent power inefficiencies. This is a particular limitation for the Department of War (DoW) when operating in austere environments, especially when using advanced artificial intelligence and machine learning (AI/ML) applications with high energy demands for both training and inference. New unconventional computing concepts, such as synthetic biological intelligence, have the potential to fundamentally address these limitations but do not yet possess the necessary training and inference capabilities for practical applications. To meet these needs, the O-Circuit program will explore methods for improving the learning, inference, and computational memory of neural tissue systems to develop an integrated biological processing unit (BPU) that establishes a new paradigm for sense-compute-act for edge applications.

Acquisition Strategy

It is anticipated that the O-Circuit acquisition strategy will use Other Transaction (OT) for Prototype awards. Abstracts for solicitation responses will be requested with an anticipated deadline of **early May 2026**. The Government will review all submitted Abstracts for technical merit and innovative solutioning. Selected proposers will be invited to provide an Oral Presentation (OP) to the Government. The Government will evaluate all Oral Presentations and anticipates that selected performers will be given an OT award with a 32-month (18-month Phase 1 base, 14-month Phase 2 option) period of performance. Abstracts are *required* to be eligible to submit an OP. DARPA encourages solutions from all responsible sources capable of satisfying the Government's needs, including large and small businesses, *nontraditional defense contractors* as defined in 10 U.S.C. § 3014, and *research institutions* as defined in 15 U.S.C. § 638.

Technical Approach

O-Circuit will develop an all-biological, converged sense-compute-act system, capable of using odorant detection information to control drone navigation tasks. The program will have two Task Areas (TAs): development of a biological processing unit (BPU) with real-time learning and inference, and persistent memory capabilities (TA1, "Architecture"); and integration of a BPU with a biologically based odorant sensing system and a drone navigation platform (TA2, "Action"). The program will focus on engineering biological computations using biological neurons. Biological processing approaches that are not centered around neural tissue (e.g., DNA computation, bacterial computation) or that are focused on neuromorphic computing solutions are not in scope.

TA1 (Architecture) will develop advanced BPU systems with the capacity to learn more complex functions and retain memory as compared to state-of-the-art neural cultures (e.g., organoid-based systems). Performers could advance techniques that increase the synaptic connectivity and plasticity within the BPU structure by such means as: optimizing cell composition and ratios, developing new nutrient exchange subsystems, and tuning metabolic processes with genetic modification and/or pharmacology. Performers could also advance the memory and learning capabilities of BPUs: for example, by updating BPU architectures with components that are based on the learning networks that are part of naturally occurring brains (e.g., in mammals and insects). The above topics are noted for example purposes only and are not intended to restrict novel ideas. To identify the most functionally promising BPU architectures, performance will be evaluated by how well the BPUs can learn, and retain that learning, in a video game simulator (e.g., Ms. Pac-Man).

TA2 (Action) will focus on developing biological 'wetware' architectures that integrate a BPU, an odorant sensor array, and drone-based navigation into a sense-compute-action system that acts on the physical world. The odorant sensor array will be biological (e.g., using olfactory receptor neurons), tailored to respond to a custom panel of operationally relevant odorants (e.g., explosives precursors, chemical agent surrogates, toxic industrial chemicals). Performers are expected to advance techniques for integrating olfactory receptor neurons or odorant receptors into BPUs, mapping their signaling to measurable BPU outputs (e.g. synaptic spikes), and constructing or training the BPU to differentiate between odorants detected by the odorant sensor array and issue navigation commands in response to the olfactory array inputs. While physical hardware will be needed to meet these objectives, proposals with significant hardware research and development efforts will be deemed out of scope. Additionally, performers will be expected to integrate the olfactory-enabled BPU with a drone navigation platform and demonstrate real-time, 2D+ physical navigation based on odorant signals.

Tentative Program Structure

O-Circuit is tentatively planned as three (3) Phases. Phase 1 (18 months) will develop and test the initial BPU architectures and the biological sense-compute-action drone systems that leverage odorant sensor arrays. Phase 2 (14 months) will refine the BPU architectures and the odorant sensor arrays to develop more advanced compute and sense capabilities. Phase 3 (10 months) will combine top performing TA1 and TA2 performers into a superteam to develop an advanced sense-compute-action system. Program events in Phases 1 and 2 will focus on Phase 3 integration planning and setting up capability demonstrations.

O-Circuit includes a Test and Evaluation (T&E) component. The T&E team will be responsible for developing a simulated game environment that will be provided for the TA1 performers at the start of Phase 1 and which will be used for assessing TA1 BPU capabilities. They will also develop a TA2 odorant testbed for assessing odorant detection and discrimination accuracy and will develop a trial environment for assessing drone navigation through odorant gradients.

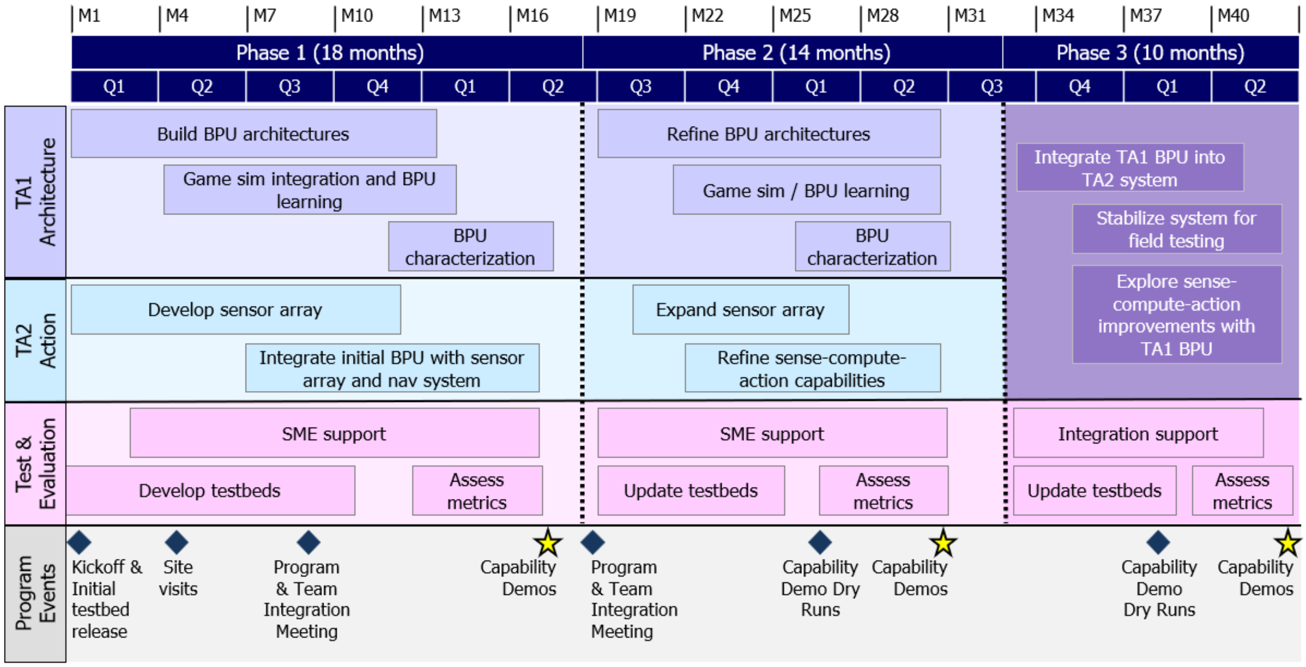

Figure 1. *Tentative O-Circuit program structure.*

Initial program proposals will only include Phase 1 and Phase 2, and teams will be allowed to propose to either a single TA or to both TAs. During Phase 2, the best-performing TA1 and TA2 efforts (based on progress towards program metrics and goals) will be eligible to submit statements of work for the Phase 3 goals (i.e., the Phase 3 superteam). At the end of each phase, performers will be required to submit results against program metrics (see table below) and conduct capability demonstrations to showcase progress on game simulator (e.g., Ms. Pac-Man) performance (TA1) and drone-based, chemical detection and navigation (TA2).

Tentative Program Metrics

	Metric	Phase 1	Phase 2	Phase 3
TA1 (Architecture)	'Ms. Pac-Man' Score	>2200*	>16,000**	
	Daily Energy Consumption†	10 mWh	10 mWh	
	Memory Persistence	24 hrs	72 hrs	
TA2 (Action)	Odorants (*n* @ 1 ppm)	50	100	
	Discrimination Accuracy	>70%	>90%	
	2D+ Gradient Descent (50m radius)	1 odorant	15 min 2 odorants	5 min 3 odorants

*Inspired from Ms. Pac-Man random score being ~300; **Representative human score ~16,000 given 2 hours of training and 5 min/match; †Nutrient consumption by the neural tissue system

ELIGIBILITY INFORMATION

Eligible Applicants

Federally Funded Research and Development Centers (FFRDCs) and Government Entities FFRDCs

FFRDCs are subject to applicable direct competition limitations and cannot propose to this solicitation in any capacity unless they meet the following conditions: (1) FFRDCs must clearly demonstrate, with specific details, that the proposed work, expertise, and facilities are not otherwise available from the private sector, and (2) FFRDCs must provide a letter on official letterhead from their sponsoring organization citing the specific authority establishing their eligibility to propose to Government solicitations and compete with industry, and their compliance with the associated FFRDC sponsor agreement's terms and conditions. This information is required for FFRDCs proposing to be awardees or subawardees. FFRDC proposals that do not include these elements may be deemed non-conforming and removed from consideration.

FFRDCs proposing as prime awardees must be able to accept an OT for Prototype agreement as the award instrument. FFRDCs that can only be funded through their existing sponsor contracts should not submit an abstract directly to this solicitation.

Government Entities

Government Entities (e.g., Government/National laboratories, military educational institutions, etc.) are subject to applicable direct competition limitations. Government entities must clearly demonstrate that the work is not otherwise available from the private sector and provide written documentation citing the specific statutory authority and contractual authority, if relevant, establishing their ability to propose to Government solicitations, and compete with industry. This information is required for Government Entities invited to submit proposals as either awardees or subawardees.

Government Entities submitting abstracts as prime awardees must be able to accept an OT for Prototype agreement as the award instrument. Government Entities that can only be funded through their existing sponsor contracts should not submit abstracts directly to this solicitation.

Authority and Eligibility

At the present time, DARPA does not consider 15 U.S.C. § 3710a to be sufficient legal authority to show eligibility. While 10 U.S.C. § 4892 (formerly 10 U.S.C. § 2539b) may be the appropriate statutory starting point for some entities, specific supporting regulatory guidance, together with evidence of agency approval, will still be required to fully establish eligibility. DARPA will consider FFRDC and Government entity eligibility submissions on a case-by-case basis; however, the burden to prove eligibility for all team members rests solely with the proposer.

Other Applicants

Non-U.S. organizations and/or individuals may participate to the extent that such participants comply with any necessary nondisclosure agreements, security regulations, export control laws, and other governing statutes applicable under the circumstances.

Organizational Conflicts of Interest (OCI)
Without prior approval or a waiver from the DARPA Deputy Director, a contractor cannot simultaneously provide scientific, engineering, technical assistance (SETA), advisory and assistance services (A&AS), or similar support and also be a technical performer. As part of the proposal, all members of the proposed team (including any potential subawardees or consultants) must affirm whether they (their organizations and individual team members) are providing SETA or similar support to any DARPA office(s) through an active award or subaward. All facts relevant to the existence or potential existence of Organizational Conflicts of Interest (OCI) must be disclosed in the Administrative and National Policy Requirements document, should the proposer be invited to submit a proposal.

If SETA, A&AS, or similar support is being or was provided to any DARPA office(s), the proposal must include in the Administrative and National Policy Requirements document:

- The name of the DARPA office receiving the support;
- The prime contract number;
- Identification of proposed team member (subawardee, consultant) providing the support;
- and
- An OCI mitigation plan.

Under this section of the proposal, the proposer is responsible for providing this disclosure with each proposal submitted. The disclosure must include the proposer's OCI status, and as applicable, proposed team member's OCI mitigation plan. The OCI mitigation plan must include a description of the actions the proposer has taken, or intends to take, to avoid, neutralize, or mitigate such conflict, prevent the existence of conflicting roles that might bias the proposer's judgment, and prevent the proposer from having unfair competitive advantage. Prior to the start of proposal evaluations, the Government will assess potential conflicts of interest based on the proposals submitted. DARPA will promptly notify the proposer if any appear to exist. The Government assessment does NOT affect, offset, or mitigate the proposer's responsibility to give full notice and planned mitigation for all potential organizational conflicts.

If, in the sole opinion of the Government after full consideration of the circumstances, a proposal fails to fully disclose potential conflicts of interest and/or any identified conflict situation cannot be effectively mitigated, the proposal will be rejected without technical evaluation and withdrawn from further consideration for award.

If a prospective proposer believes a conflict of interest exists or may exist (whether organizational or otherwise) or has questions on what constitutes a conflict of interest, the proposer should send his/her contact information and a summary of the potential conflict via the specific email address identified in this solicitation before time and effort are expended in preparing a proposal and mitigation plan.